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Travelling to Bundesliga matches: the carbon footprint of football fans

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ABSTRACT

The purposes of this study were to estimate the carbon footprint caused by football fans travelling to Bundesliga matches (first division) in Germany in the 2018/19 season, to analyse determinants of seasonal carbon footprint, and to identify fan clusters based on travel behaviour. A nationwide online survey of football fans was conducted (n = 539) asking respondents to report their match-related travel behaviour. The average seasonal carbon footprint of a Bundesliga fan amounted 311.1 kg of carbon dioxide equivalent emissions (CO_2-e), with car travel accounting for 70% of the emissions. The aggregate carbon footprint of all fans for the whole Bundesliga season was 369,765.2 t CO_2-e. Buying out of these carbon emissions would cost over €9.2 million in total. The regression results revealed that club membership and commitment to favourite club significantly increased fans' carbon footprint. The choice of favourite Bundesliga club also predicted carbon footprint, with fans of FC Bayern Munich and RB Leipzig producing a significantly higher carbon footprint than fans of Borussia Dortmund. The segmentation of Bundesliga fans by their travel behaviour identified three distinct clusters: Devoted travellers (19%), home fans (30%) and casual visitors (51%). These clusters differed significantly in terms of emissions per kilometre travelled, car use versus public transport, club membership, fan commitment, environmental consciousness, education and age. The findings have practical implications for policy makers, Bundesliga officials and club authorities, and can serve as a basis for initiatives to reduce carbon dioxide emissions in professional team sports.

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Carbon dioxide emissions; climate change; environmental consciousness; pro-environmental behaviour; professional sport

Introduction

The global scientific community has stressed that planet Earth is facing a climate emergency (Ripple, Wolf, Newsome, Barnard, & Moomaw, 2020). The Intergovernmental Panel on Climate Change (IPCC) reported that human activities have already caused approximately 1°C of global warming above pre-industrial levels (IPCC, 2018). The environmental consequences are alarming: Temperatures of the atmosphere and oceans have warmed, snow and ice masses have melted, sea levels have risen, and the

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concentration of greenhouse gases in the atmosphere has increased (IPCC, 2013). These critical developments require environmental actions to be taken in several sectors of human life to combat anthropogenic climate change. One such sector is travel and transport which makes significant contributions to the overall amount of greenhouse gas emissions and hence to climate change (IPCC, 2018).

Sport is one area where people travel a lot – not only for actively practising different sports (Thormann & Wicker, 2021; Wicker, 2019), but also as spectators (Cooper, 2020). Thus, travel activities related to sport tourism and sport events contribute to greenhouse gas emissions and the overall environmental impact of sport (McCullough, Orr, & Watanabe, 2019). Existing research examining the environmental impact of sport spectators has focused on one-off events (Collins, Flynn, Munday, & Roberts, 2007; Scrucca, Severi, Galvan, & Brunori, 2016) and collegiate sports (Cooper, 2020; Dolf & Teehan, 2015). The emissions caused by fans of professional team sport leagues have not been studied yet, although these leagues draw sizeable audiences. Since fans' attendance and travel patterns vary over the course of a season, a season-wide assessment is necessary.

The focus of this study is on the German Football Bundesliga, the highest professional football league in Germany, which attracted a total of 13.3 million fans in the 2018/19 season (Transfermarkt, 2019). These fans travelled short and long distances across the country to attend Bundesliga matches, indicating that substantial greenhouse gas emissions might be associated with the Bundesliga. According to their websites, some Bundesliga clubs actively take measures to reduce emissions and publicly commit to climate protection (e.g. Mainz 05, 2021; TSG Hoffenheim, 2020; VfL Wolfsburg, 2020), while others do not make any statements about the promotion of environmentally sustainable practices (e.g. FC Bayern Munich, RB Leipzig, Bayer Leverkusen, Eintracht Frankfurt, FC Schalke). Accordingly, the task force of the German Football League (DFL) recently defined the objective of reinforcing the commitment to ecological sustainability criteria in the Bundesliga until 2030 (DFL, 2021). To achieve this objective and develop measures, it is crucial for league officials and clubs to understand the extent and drivers of environmental impacts.

The purpose of this study is to analyse the carbon footprint caused by football fans travelling to Bundesliga matches (first division) in Germany. This research focusses on the 2018/19 season which is the last season that was finished with regular match attendances before the COVID-19 pandemic. This study advances three main research questions: (1) what is the seasonal carbon footprint of football fans in the 2018/19 Bundesliga season? (2) What factors contribute to fans' carbon footprint? And (3) into what groups can Bundesliga fans be segmented based on their travel behaviour? The research questions are answered using data from a comprehensive online survey on the travel behaviour of football fans. The empirical findings will highlight to what extent fan travel contributes to climate change and demonstrate which factors influence fans' seasonal carbon footprint.

Theoretical framework and literature review

Concept and measurement of carbon footprint

The carbon footprint can be defined as 'the quantity of greenhouse gases expressed in terms of carbon dioxide equivalent emissions, emitted into the atmosphere by an

individual, organisation, process, product, or event from within a specific boundary' (Pandey, Agrawal, & Pandey, 2011, p. 138). Conceptually, the carbon footprint is a global warming indicator as it includes emissions of several greenhouse gases (Pandey et al., 2011). In a carbon footprint analysis, the emissions of these greenhouse gases are normalised to the global warming potential of carbon dioxide and converted into carbon dioxide equivalents (CO_2-e) which is the carbon footprint's unit of measurement. Thus, the carbon footprint reflects the global warming potential of the greenhouse gases relative to that of CO_2 (Franchetti & Apul, 2013; Pandey et al., 2011).

Franchetti and Apul (2013) have specified three boundaries to conduct a carbon footprint analysis. The organisational boundary defines the entity for which the carbon footprint is estimated (e.g. for an individual, a nation, or an organisation), whereas the temporal boundary determines the time period in which all activities are included in the footprint analysis. The operational boundary distinguishes between three scopes of emissions to avoid double counting. Scope 1 emissions are direct emissions that can be caused by mobile sources, such as vehicles (transport, travelling, fuel consumption), and by stationary combustion for producing electricity, heat, or steam (Franchetti & Apul, 2013). Scope 2 and 3 emissions include all indirect emissions that are either linked to the consumption and purchase of energy in the form of electricity, steam or water (Scope 2) or that occur during the whole product life cycle from resource acquisition to waste management (Scope 3; Franchetti & Apul, 2013; Pandey et al., 2011). The present research focuses on direct emissions (operational boundary) caused by football fans travelling to the matches of Bundesliga in Germany (organisational boundary) in the 2018/19 season (temporal boundary).

Empirical evidence on the carbon footprint of sport events

Scholars have investigated the carbon footprint of different types of sport events, including one-off events and sport leagues with a year-long season. The focus was on one-off events such as the 2004 Wales Rally (Jones, 2008), the 2007 Tour de France stages held in the UK (Collins, Munday, & Roberts, 2012) and the 2014 World Orienteering Championship (Scrucca et al., 2016). Regarding sport events with league character, existing research has studied ten university sport teams in Canada over two seasons (Dolf & Teehan, 2015) and the home matches of an American football team at a university in the United States over four seasons (Cooper, 2020). Most closely related to the present research context, Dosumu, Colbeck, and Bragg (2017) estimated a total carbon footprint of 56,237 t CO_2-e caused by fan travel in eight English semi-professional and non-professional football leagues, resulting in an average emission of 4.7 kg CO_2-e per fan. Concerning transportation modes, Dosumu et al. (2017) reported that most fans travelled by car (67%) to football matches in lower football divisions, with car travel and public transport producing the major share of emissions.

Although the body of research examining the environmental impacts of sport events has added valuable insights, the literature is not without shortcomings. First, most studies were descriptive, estimating aggregate or average carbon footprints. In spectator sports, determinants of an individual's carbon footprint and different clusters of fans/consumers affecting the environment have not been studied yet. Second, previous sport event research focused on one-off sport events rather than whole seasons including home

and away matches of all teams of a sport league. The one existing study only assessed semi-professional and non-professional sports. However, professional sport leagues tend to have a much larger number of fans influencing their overall environmental impact. Also, such national leagues have a larger geographical reach which might affect fans' travel patterns and transport modes. Since fans vary in their attendance and travel patterns in the sense that every fan might attend a different combination of home and away matches and might use different transport modes, it is important to assess all matches of a season and not only one match. The present study aims to address both gaps. It attempts to contribute to the literature by estimating the seasonal carbon footprint of football fans in the first division of professional football in Germany. Furthermore, it takes an analytical approach to examine determinants of seasonal carbon footprint and identify fan clusters based on travel behaviour.

Factors affecting a fan's seasonal carbon footprint

This section advances potential determinants of carbon footprint including environmental consciousness, fan commitment, and club membership. First, environmental consciousness reflects an individual's general attitudes towards the environment (Preisendörfer, 1999). Environmental consciousness captures several environmental aspects and personal orientations in terms of environmental damage, knowledge about environmental problems, emotional reactions, attitudes to political environmental measures, and general value orientations. The concept encompasses a cognitive component (understanding of the endangerment resulting from environmental damage), a conative component (willingness to act environmentally friendly) and an affective component (emotional reactions to environmental problems).

The theory of reasoned action suggests that human beings act rationally by making systematic use of the information available to them. In the process of acting rationally, attitudes determine an individual's actual behaviour (Ajzen & Fishbein, 1980). Applying this theory to the present research context, environmental consciousness (attitude towards the environment) is expected to predict an individual's pro-environmental behaviour (actual behaviour). The general assumption that environmental consciousness is positively associated with pro-environmental behaviour is also suggested by theoretical models of environmental behaviour (Diekmann & Preisendörfer, 2003; Kollmuss & Agyeman, 2002).

However, previous research examining this relationship in sport came to opposing results. On the one hand, environmental consciousness was positively related to the pro-environmental behaviour of participants in individual sports (Wicker, 2019) and team/racket sports (Thormann & Wicker, 2021) as well as sport fans (Casper, Pfahl, & McCullough, 2014). On the other hand, several researchers were not able to find a significant contribution of environmental consciousness on the pro-environmental behaviour of sport students (Casper & Pfahl, 2012), snow sport tourists (Wicker, 2018), nature sports participants (Wicker, 2019) and sport spectators (McCullough & Cunningham, 2011).

This discrepancy between environmental consciousness and pro-environmental behaviour is referred to as the 'environmental value-action gap' (Blake, 1999, p. 268). The low-cost hypothesis (Diekmann & Preisendörfer, 2003) aims to explain this gap, outlining that environmental consciousness only leads to pro-environmental behaviour in

low-cost situations (e.g. separating waste). When the costs in terms of time, money, or convenience that are associated with pro-environmental behaviour are perceived as too high, individuals do not act environmentally friendly. For instance, travelling with environmentally friendly modes of transport can represent a high-cost situation due to the perceived benefits and convenience of cars and air travel. Despite some contradicting empirical results, the first hypothesis follows the theoretical tenets and assumes that environmentally conscious individuals act environmentally friendly and cause lower levels of carbon emissions:

H1: Environmental consciousness is negatively associated with seasonal carbon footprint.

Fans' connection with their favourite football club might also affect an individual's carbon footprint. Since the theoretical environmental literature is silent about such factors and their association with emissions, we draw on other theoretical approaches of fan research. The commitment, or loyalty, of a fan to his/her favourite football club could serve as a predictor of a fan's carbon footprint. Fan loyalty is characterised as a highly developed attitude that is persistent and resistant to change, while at the same time creating biases in cognitive processing and guiding behaviour (Funk & James, 2001). This concept has especially been studied in conjunction with sport sponsorship (e.g. Silva, 2020). However, fan identity with a club was also found to be a significant factor of fan-related behaviour, including match attendance (Laverie & Arnett, 2000). Team loyalty was even found to be the most important predictor for spectators' desire to attend a live sporting event (Wakefield & Sloan, 1995). Hence, it is assumed that fans who are highly committed to their favourite club attend more matches and are willing to travel further to see their team play, consequently resulting in a higher carbon footprint. The second hypothesis captures this relationship:

H2: Fan commitment is positively associated with seasonal carbon footprint.

Club membership also demonstrates a strong emotional connection to a football club. Heere and James (2007) point out that club membership reinforces loyalty to a sports team. In line with the second hypothesis, it is assumed that fans who are a member of their favourite club show a strong loyalty and, therefore, regularly attend matches causing a higher carbon footprint. The third hypothesis reflects this aspect:

H3: Club membership is positively associated with seasonal carbon footprint.

Method

Data collection

Primary data were collected using an online survey of football fans of the first division of the Bundesliga in the 2018/19 season. The survey addressed football fans of Bundesliga clubs who had visited at least one match in the 2018/19 season and who had their primary residence in Germany. This geographical restriction was necessary because emission factors of transport modes only apply to Germany (Federal Environmental Office, 2018b). The online survey was provided on the platform Social Science Survey (www.soscisurvey.de).

Data were collected using a convenience sampling strategy. The link to the online survey was posted on several websites of news and sports magazines, on Facebook, and in online discussion forums about the Bundesliga. In addition, the survey was forwarded to several fan projects of all Bundesliga clubs that were asked to share the survey link within their fan community. The survey was online shortly after the final match of the 2018/19 Bundesliga season from 20th June to 8th July 2019 to assure that fans were able to recall the Bundesliga matches they had visited and the transport modes they had used.

This sampling procedure resulted in 1370 clicks on the link, 741 individuals commenced the survey, and 563 completed it. During the data cleaning process, 24 cases were excluded for various reasons, including respondents not having visited a Bundesliga match in the 2018/19 season or stating invalid or foreign postcodes. Valid German postcodes were crucial for the calculation of the fans' travel distance to the matches and the application of the German emission factor table (Federal Environmental Office, 2018b). After checking the respondents' answers for plausibility and internal validity, a final sample of $n = 539$ fans remained for the empirical analysis.

Questionnaire and variables

The online survey started with an introduction informing respondents about the anonymity of the survey, the confidential treatment of data, and the use of data only for scientific purposes. The survey consisted of three sets of questions, including respondents' fan profile, travel behaviour, and socio-demographics. Table 1 gives an overview of the variables included in this study.

First, fans were asked to state their favourite first division team (*Fan_CLUB)*, if they were a member of their favourite club (*Membership*), and about their commitment to their favourite club. Commitment was assessed using the psychological commitment to

Table 1. Overview of variables and summary statistics.

Variable	Description	Mean	SD
CF_season	Individual seasonal carbon footprint (in g CO_2-e)	311,134.89	494,838.40
LN_CF	Natural logarithm of CF_season	11.11	2.84
Membership	Club membership (1 = yes, 0 = no)	0.547	–
Fan_commitment	Psychological commitment to team (mean index; 1 = not committed; 7 = highly committed; items see Table A1)	6.30	0.69
Home_matches	Number of home matches visited	7.93	6.26
Away_matches	Number of away matches visited	3.31	4.34
Env_conscious	Environmental consciousness index (mean index; 1 = not environmentally conscious; 5 = highly environmentally conscious; items see Table A2)	3.67	0.83
Male	Respondent's gender (1 = male; 0 = female)	0.915	–
Age	Respondent's age (in years)	32.65	11.04
Age_sq	Age squared (=Age*Age)	1,187.48	863.02
Edu_low	Highest educational level is below A-levels (1 = yes; 0 = no)	0.251	–
A-levels	Highest educational level is university entry qualification (1 = yes; 0 = no)	0.258	–
University	University or university of applied sciences degree (1 = yes; 0 = no)	0.492	–
Income	Individual monthly net income in € (1 = 0–500; 2 = 501–1,000; 3 = 1,001–1.500; … .; 9 = over 4000)	4.53	2.13
Fan_CLUB	Respondent's favourite Bundesliga club (18 Dummies coded 1 = yes, 0 = no; Figure 1)	–	–

team scale (Mahony, Madrigal, & Howard, 2000). The scale consisted of fourteen items which were measured on a 7-point Likert scale (see Table A1 for the concrete wording of items and their descriptive statistics). Items 1, 3, 8 and 14 were reverse-coded to identify and remove respondents who simply clicked through the survey. Construct reliability was assessed using Cronbach's alpha which was 0.81 and, therefore, exceeded the suggested threshold of 0.8 (Hair, Black, Babin, & Anderson, 2010). As construct reliability was satisfactory, the fourteen items were converted into an index (*Fan_commitment*) by adding up the items and dividing the sum by fourteen.

Second, fans' travel behaviour in relation home and away matches was assessed. For home matches, respondents were asked to state how many of them (*Home_matches*) they had visited in the 2018/19 Bundesliga season, which distance they travelled to get to the home stadium ground, and which mode of transport they typically used. As fans might have used more than one mode of transport on their journey to a home match (e.g. fans might have travelled parts of the distance by car and parts by local train), they were asked to indicate all transport modes used. The survey listed the same modes of transport that are included in the emission factor table of the German Federal Environmental Office (2018b): *Coach, car, long-distance train, local train, public bus*, *subway/tram,* and *plane*. Two answer options were added for fans who biked or walked to the stadium (*bicycle, walk*).

In the away match section, a list of all seventeen potential away matches was presented. Respondents were asked to click on the away matches they attended. For every single away match, participants could state which of the above modes of transport they used to cover the largest distance to the respective away match ground. The assessment of transport modes and locations in surveys is feasible as this is the type of information respondents are able to provide; they are likely unaware of indirect or lifecycle emissions (Mathez, Manaugh, Chakour, El-Geneidy, & Hatzopoulou, 2013).

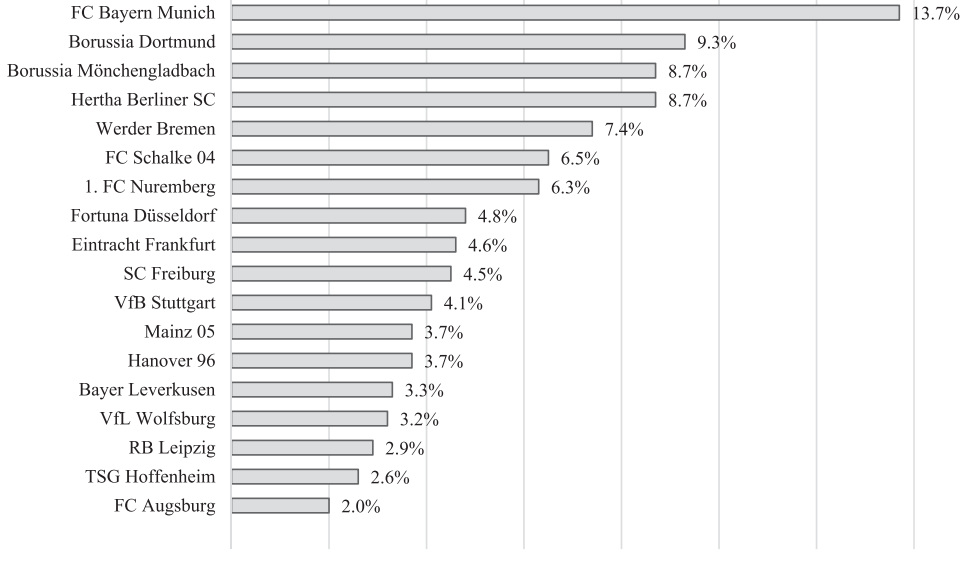


Figure 1. Bundesliga fans' favourite club (share of respondents in %).

Respondents' environmental consciousness was assessed using an established scale developed by Diekmann and Preisendörfer (2003), which has been applied in previous carbon footprint research (Thormann & Wicker, 2021; Wicker, 2018, 2019). The scale includes affective (items 1–3), cognitive (items 4–6), and conative statements (items 7–9) capturing respondents' general environmental consciousness on a 5-point Likert scale (see Table A2 for items and descriptive statistics). The nine statements were presented in randomised mixed order to respondents to avoid that the three components of environmental consciousness were tested one after the other. Cronbach's alpha was 0.89, supporting construct reliability (Hair et al., 2010). The nine items were converted into an index (*Env_conscious*) representing fans' environmental consciousness.

The survey concluded with socio-demographic questions about respondents' post-code, age (*Age*), gender *(Male*), educational level (*Edu_low; A-levels; University*) and income (*Income*). Age squared (*Age_sq*) was computed to capture possible non-linear effects of age on carbon footprint.

Calculation of carbon footprint

Fans' travel-related carbon footprint was estimated based on the travel distances to the stadium grounds and the modes of transport used to attend the matches – similar to existing carbon footprint research (Thormann & Wicker, 2021; Wicker, 2018, 2019). The travel distances for each fan and each match were calculated based on the stated distance between the respondent's home residence post code and the respective stadium grounds. This procedure was employed for 6056 match attendances of the 2018/19 Bundesliga season (4274 home and 1782 away matches).

The travel distances that fans covered with the respective modes of transport were converted into carbon dioxide equivalent emissions (CO_2-e) using the emission factor table provided by the German Federal Environmental Office (2018b; Table A3). This table includes emission factors for each mode of transport and relies on their average degree of capacity utilisation in Germany as evident in extensive transportation research. The emission factors capture direct emissions (Scope 1) and reflect the CO_2-equivalent emissions per person per kilometre in Germany for the relevant modes of transport. Biking and walking were assigned an emission factor of zero as both ways of travelling do not produce any direct emissions.

The carbon footprint for each fan was calculated by multiplying the distance to the stadium ground with the emission factor of the respective mode of transport. Since the survey captured one-way distances, CO_2-equivalent emissions were multiplied by the factor 2 to take return trips into account. The CO_2-equivalent emissions for each match attended were added up, resulting in a fan's total carbon footprint for the 2018/19 Bundesliga season (*CF_season)*. The carbon footprint reflects fans' variations in attendance and travel patterns over the course of a season.

Empirical analysis

The empirical analysis included three steps. First, descriptive statistics were provided to give an overview over the sample structure and fans' seasonal carbon footprint – also differentiated by club. Moreover, the frequency of transport modes and their contribution

to the total carbon footprint of Bundesliga fans were reported. Furthermore, the aggregate carbon footprint for all fans for the whole season was estimated. The aggregation is based on the total number of fans attending Bundesliga matches in the 2018/19 season, which is 13,300,905 (Transfermarkt, 2019). This figure is an accumulated number of the attendances of all Bundesliga matches counting every match attendance separately. Put differently, fans visiting multiple matches during the season are included multiple times in this figure. The aggregate carbon footprint of all fans for the whole season was calculated by multiplying the average carbon footprint per fan per match with this total match attendance figure. In addition, the seasonal aggregate is converted into monetary equivalents using a carbon emission trading price of €25 per ton CO_2-e in Germany (Federal Government, 2019).

Second, a linear regression model was estimated to identify factors associated with seasonal carbon footprint. Since the dependent variable (*CF_season*) was positively skewed, its natural logarithm (*LN_CF*) was computed to move the variable closer to the normal distribution. The other variables from Table 1 were entered as independent variables in the regression model, except for the number of home and away matches visited as they were used in the carbon footprint calculation. Hence, the empirical analysis reflects previous research on the factors associated with pro-environmental behaviour and carbon footprint (e.g. Bamberg & Möser, 2007; Thormann & Wicker, 2021). All independent variables were checked for multicollinearity using variance inflation factors and correlation analyses. Since all correlations were below the suggested threshold of 0.8 and all variance inflation factors were below the threshold of 10 (except for *Age* and *Age_sq*), there were no indications of multicollinearity (Hair et al., 2010).

In a third step, a two-step cluster analysis was conducted to identify different segments of Bundesliga fans based on their travel behaviour in terms of carbon footprint and the number of home and away matches visited. Analyses of variances (ANOVA) were run to test whether the identified clusters showed significant differences in terms of fan profile, environmental consciousness, and socio-demographic characteristics. A post hoc test was employed to examine which clusters significantly differed from each other. For all statistical tests, an α-level of 0.05 was used.

Results and discussion

Summary statistics

Table 1 reports the descriptive statistics. Starting with the socio-demographic characteristics, 92% of respondents were male and on average 32.7 years old. Altogether, 26% of respondents held a university entry qualification and 49% graduated from university or university of applied sciences. Fans' average personal monthly net income was 4.5 on the income scale, reflecting a monthly net income of about €2000.

Respondents showed a rather high commitment to their favourite team on the psychological commitment to team scale (M = 6.30 on a seven-point scale). Figure 1 shows the breakdown of fans by clubs, indicating that the highest share in the sample (13.7%) were fans of FC Bayern Munich. More than half of respondents (55%) were a member of their favourite club. Fans' environmental consciousness was 3.67 on a five-point scale. The

average Bundesliga fan in this sample attended 11.2 out of 34 matches in the 2018/19 season, including 7.9 home matches and 3.3 away matches.

Seasonal carbon footprint

The average seasonal carbon footprint per fan is 311.1 kg CO_2-e, reflecting the variety of attendance and travel patterns over the course of a season. This seasonal carbon footprint of Bundesliga fans was 66 times higher than the carbon footprint of football fans across eight English non-professional football divisions (4.7 kg CO_2-e; Dosumu et al., 2017) where travel distances were much shorter than in a professional national league like the Bundesliga. Compared with the annual emissions of a person living in Germany which amounted to 10.4 t CO_2-e (Federal Environmental Office, 2018a), the carbon footprint of Bundesliga-related travel made up 3.0% of the overall emissions. Figure 2 visualises the average carbon footprints by club. It shows that the carbon footprint of fans of FC Bayern Munich (673.8 kg CO_2-e) was more than twice as high as the sample mean, followed by fans of RB Leipzig (387.7 kg CO_2-e), 1. FC Nuremberg (360.7 kg CO_2-e), and Borussia Mönchengladbach (353.7 kg CO_2-e).

Figure 3 summarises how frequently each mode of transport was used by fans to travel to Bundesliga matches and how much each transport mode contributed to total emissions. The most frequently used mode of transport was the private car which fans used for 36.1% of the journeys to Bundesliga matches. This finding is similar to previous research showing that football fans preferred cars to attend football matches (Dosumu et al., 2017). Car travel dominated the carbon footprint as it generated 70.3% of the total emissions although fans travelled to only one third of all matches by car. Domestic air travel showed similar effects on seasonal carbon footprint. While fans only travelled to 0.7% of the matches by plane, air travel accounted for 5% of seasonal carbon footprint. Especially flights to away matches in Berlin and Munich were responsible for more than half of the air travel emissions in the present study (54%). Travelling by train and local public transport caused relatively low emissions compared to the usage frequency.

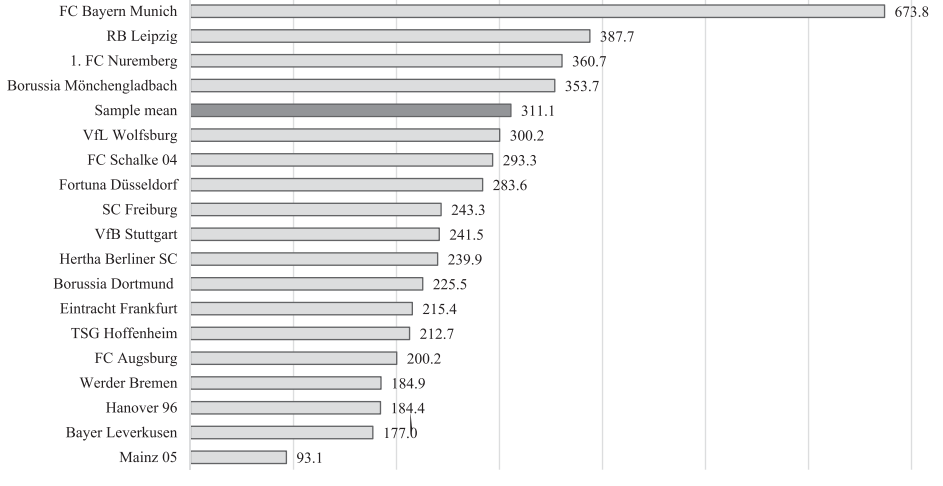


Figure 2. Seasonal carbon footprint by favourite Bundesliga team (mean in kg CO_2-e).

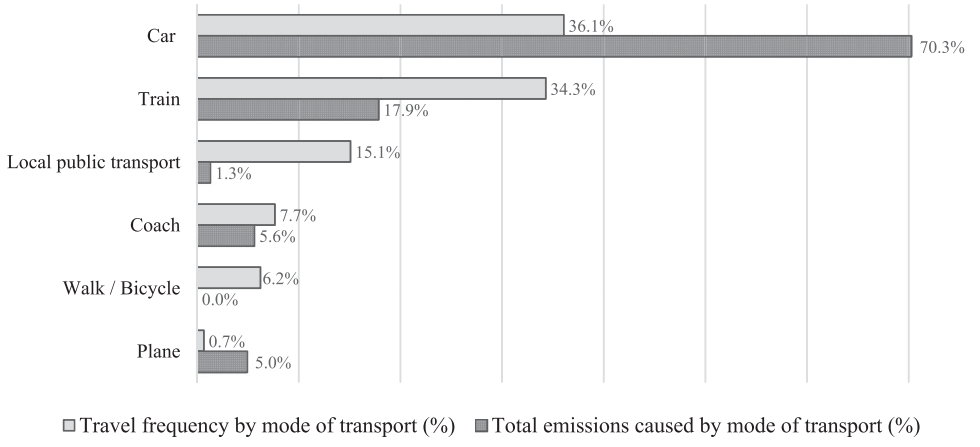


Figure 3. Bundesliga fans' transport mode choices and the transport modes' relative contribution to seasonal carbon footprint (in %).

The total environmental impact of fans travelling across Germany on 34 matchdays to the matches of their favourite teams becomes evident when estimating the aggregate carbon footprint that is caused by all fans over the whole season. Based on the average seasonal carbon footprint of a Bundesliga fan of 311.1 kg CO_2-e and the average number of matches visited of 11.2 matches, the average carbon footprint amounts to 27.8 kg CO_2-e per fan per match. Multiplying this average with the total match attendances of 13,300,905 fans generates a total carbon footprint of approximately 369,765.2 t CO_2-e over the whole season. Buying out of carbon emissions resulting from fan travel to Bundesliga matches would cost over €9.2 million in total.

Factors associated with seasonal carbon footprint

Table 2 summarises the results of the log-linear regression model analysing the determinants of fans' carbon footprint. The model was significant and its predictors were able to explain 22% of the variance in seasonal carbon footprint. In contrast to previous environmental research (e.g. Bamberg & Möser, 2007; Thormann & Wicker, 2021; Wicker, 2019), socio-demographic variables had no significant effect on carbon footprint.

The regression analysis revealed a negative, but insignificant effect of environmental consciousness on carbon footprint. Consequently, the assumption that higher levels of environmental consciousness lead to a lower carbon footprint does not hold and H1 must be rejected. This finding supports the existence of an environmental value-action gap (Blake, 1999), suggesting that fans who considered themselves environmentally conscious did not demonstrate environmentally friendly behaviour in the form of a lower carbon footprint. This discrepancy between environmental consciousness and pro-environmental behaviour was also found among sport spectators in collegiate athletics (McCullough & Cunningham, 2011) and active sport participants (e.g. Casper & Pfahl, 2012; Wicker, 2018, 2019).

The environmental value-action gap can be explained by the low-cost hypothesis, assuming that individuals only act environmentally friendly in low-cost situations

Table 2. Regression analysis for seasonal carbon footprint (LN_CF) of Bundesliga fans.

	Coeff.	*t*
Constant	8.956	4.644***
Membership	1.627	6.400***
Fan_commitment	0.566	3.155**
Env_conscious	−0.195	−1.385
Male	−0.202	−0.479
Age	−0.097	−1.416
Age_sq	0.001	1.053
Edu_low	REF	REF
A-levels	0.129	0.375
University	0.032	0.108
Income	0.048	0.673
Fan_BorussiaDortmund	REF	REF
Fan_FCBayernMunich	1.273	2.632**
Fan_RBLeipzig	2.214	2.855**
Fan_BayerLeverkusen	0.052	0.072
Fan_BorussiaMönchengladbach	0.161	0.302
Fan_VfLWolfsburg	0.770	1.056
Fan_EintrachtFrankfurt	0.863	1.350
Fan_WerderBremen	0.728	1.313
Fan_TSGHoffenheim	0.805	1.007
Fan_FortunaDüsseldorf	0.125	0.197
Fan_HerthaBerlinerSC	−0.113	−0.210
Fan_Mainz05	−1.619	−2.348*
Fan_SCFreiburg	0.264	0.407
Fan_FCSchalke	0.681	1.161
Fan_FCAugsburg	0.797	0.911
Fan_VfBStuttgart	−0.668	−1.001
Fan_Hanover96	0.516	0.751
Fan_1.FCNuremberg	1.088	1.881
R^2	0.219	
F	5.536	
p	<0.001***	

Note: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; displayed are the non-standardised coefficients; REF = reference category.

(Diekmann & Preisendörfer, 2003). In high-cost situations, individuals perceive the costs in terms of time, money, and convenience of using the more environmentally friendly alternative as too high (Diekmann & Preisendörfer, 2003). Hence, travelling to Bundesliga matches seemed to represent a high-cost situation to fans. The perceived costs might be related to the mode of transport. The most frequently used transport mode was the car, causing the majority of emissions. Car travel is often considered convenient and time-efficient (Becken, 2004). Fans might have regarded the costs of using other, more environmentally friendly modes of transport as too high. Collectively, the present findings support the notion that sport-related travel is a high-cost situation where environmental consciousness does not translate into pro-environmental behaviour.

Fans' commitment to their favourite club was significantly and positively associated with carbon footprint, confirming H2. When the score on the psychological commitment to team scale increases by one unit, carbon emissions increase by 56.6%. This finding supports the assumption that fans who are strongly committed to their favourite club produce higher emissions since they attend more matches and are willing to travel further to see their team play. This finding is in line with existing research reporting that loyal fans have a much higher likelihood to attend matches of their favourite team (Laverie & Arnett, 2000; Wakefield & Sloan, 1995).

Club membership was a significant positive predictor of seasonal carbon footprint, supporting H3. Club members had a carbon footprint that was about four times higher (409%) than those of non-members. Club membership reflects an emotional connection to a sport team (Heere & James, 2007), leading to a higher willingness to attend matches featuring the favourite club. Moreover, members of professional sport clubs often enjoy a pre-emptive right at ticket sales. In the Bundesliga, clubs guarantee their members the right to purchase match tickets first before non-members can order tickets. At some Bundesliga clubs where ticket demand is particularly high only members have the opportunity to purchase tickets. Thus club members have better access to tickets and might, therefore, attend more matches resulting in a higher carbon footprint.

The regression analysis revealed that the favourite club significantly affected the carbon footprint of Bundesliga fans. Fans of FC Bayern Munich had a significantly higher carbon footprint than fans of Borussia Dortmund who served as the reference category. This finding can be partially explained by the location of clubs. With 469.7 km, the average distance between the home stadium ground of FC Bayern Munich and the other Bundesliga stadiums was rather large compared to the Bundesliga average (376.9 km). This might explain higher away match emissions of fans of FC Bayern Munich, assuming that they live in or close to the area of Munich. Furthermore, FC Bayern Munich is the most successful club in Bundesliga history attracting fans from all over Germany. These fans from different parts of Germany have to travel large distances to Bayern Munich's home stadium ground in comparison to fans of clubs that have a rather regional fan base like Bayer Leverkusen, TSG Hoffenheim, or SC Freiburg. This explanation is supported by the structure of the sample, indicating that a large share of supporters of FC Bayern Munich had their home residence relatively far away from Munich, especially in the Western part of Germany.

RB Leipzig fans also had a significantly higher carbon footprint than those of Borussia Dortmund. The city of Leipzig is located in the Eastern part of Germany with a low density of Bundesliga clubs, causing larger travel distances to away matches for RB Leipzig fans. In comparison, the reference club Borussia Dortmund is based in the Western part of Germany where many other Bundesliga clubs have their home stadium ground. On the contrary, fans of Mainz 05 had a significantly lower carbon footprint than fans of Borussia Dortmund. Mainz 05 is also known for having a rather regional fan base from the Rhineland. With about 304 km on average, Mainz 05 fans had a relatively small travel distance to the other 17 Bundesliga stadiums.

Fan clusters based on travel behaviour

The cluster procedure based on fans' travel behaviour yielded three clusters which were labelled *devoted travellers*, *home fans* and *casual visitors*. *Devoted travellers* attended about three quarters of Bundesliga matches of their team and, therefore, had a significantly higher average carbon footprint (1047.2 kg CO_2-e) than fans in the other clusters. *Home fans* visited numerous home matches, but only a few away matches, resulting in a lower average carbon footprint (170.3 kg CO_2-e). Fans in the *casual visitors* cluster attended home and away matches only occasionally, causing a lower carbon footprint (121.3 kg CO_2-e).

Further differences in travel behaviour between the clusters emerged. *Casual visitors* have the highest emission level per km travelled, which is significantly different from *home fans*, but not from *devoted travellers*. This finding originates from the mode of transport that is used for covering the largest distance to the match location. Here, *devoted travellers* have the highest share of car use and the lowest of public transport, which is significantly different from *home fans*. Likewise, *casual visitors* have significantly higher (car use) and lower (public transport use) shares than *home fans*. Collectively, *devoted travellers* and *casual visitors* have similar transport patterns despite attending a different number of matches.

Regarding other fan characteristics, *devoted travellers* scored significantly higher on fan commitment and were significantly more likely to be a member of their favourite club, indicating their passion and allegiance to the team. *Devoted travellers* also differ from fans in the other clusters in terms of environmental consciousness, age, and educational level. Specifically, *devoted travellers* showed significantly lower levels of environmental consciousness than fans in the other two clusters. Hence, fans in the cluster with the highest emission levels were less environmentally conscious. This finding supports the general assumptions that low levels of environmental consciousness result in low levels of pro-environmental behaviour (Diekmann & Preisendörfer, 2003).

Devoted travellers were also significantly younger than *home fans.* Thus fans in the youngest cluster had the lowest environmental consciousness and produced the highest emissions. Supporting their team at home and away matches all over the country seems to be an essential part of the fan identity of this younger fan group and results in higher emissions. This finding is surprising as recent movements like *Fridays for Future* might signal that younger people care more about environmentally friendly behaviour. On the contrary, respondents in the *home fans* cluster were significantly

Table 3. Results of the cluster analysis, ANOVA, and post hoc test (displayed are the mean values).

	Cluster 1: Devoted travellers	Cluster 2: Home fans	Cluster 3: Casual visitors	*F*
Number of cases	102	161	276	
Share of cases	19%	30%	51%	
Cluster variables				
CF_season	1,047,225.89^{bc}	170,270.94^{a}	121,271.89^{a}	289.514***
Home_matches	14.02^{c}	13.62^{c}	2.36^{ab}	1,345.923***
Away_matches	10.89^{bc}	2.01^{ac}	1.26^{ab}	684.197***
Fan characteristics				
Emissions per km travelled (in g CO_2-e)	85.31	77.20^{c}	89.08^{b}	4.051*
Car use (in %)	44.2^{b}	25.8^{ac}	42.8^{b}	9.912***
Public transport (in %)	38.5^{b}	60.0^{ac}	47.0^{b}	8.442***
Fan_commitment	6.66^{bc}	6.30^{a}	6.16^{a}	21.444***
Membership	0.85^{bc}	0.69^{ac}	0.35^{ab}	56.842***
Env_conscious	3.47^{bc}	3.73^{a}	3.71^{a}	3.846*
Male	0.89	0.89	0.93	1.469
Age	30.88^{b}	34.99^{ac}	31.93^{b}	5.591**
Edu_low	0.34^{c}	0.24	0.22^{a}	3.017*
A-levels	0.22	0.26	0.27	0.615
University	0.44	0.50	0.51	0.661
Income	4.38	4.65	4.51	0.521

Note: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; post hoc test results (Bonferroni): ^{a}significantly different from cluster 1; ^{b}significantly different from cluster 2; ^{c}significantly different from cluster 3.

older than those in the other two clusters, suggesting that older age can represent a restricting factor for frequent travelling and travelling longer distances.

The *devoted travellers* cluster contains a significantly higher share of fans with an educational degree below A-levels than the *casual visitors* cluster. This finding indicates that *casual visitors* who produced the smallest emissions were better educated than *devoted travellers* who caused the highest carbon footprint. This result supports at least partially the assumption that higher levels of education contribute positively to pro-environmental behaviour (Kollmuss & Agyeman, 2002) (Table 3).

Conclusion

This study estimated the seasonal carbon footprint of football fans in the 2018/19 Bundesliga season, analysed its determinants, and identified three fan clusters based on travel behaviour. The findings provided novel insights into the emissions of fans. The results showed that a high seasonal carbon footprint is driven by club membership, team commitment, and the choice of the favourite club. The cluster analysis revealed that younger, committed fans were in the cluster with the highest emissions and the lowest levels of environmental consciousness and education, while favouring car use over public transport.

This work adds to the body of literature on environmental sustainability in sport since it was (to the best knowledge of the authors) the first study to estimate the carbon footprint of sport spectators in a major professional sport league over the period of a whole season. Such a seasonal carbon footprint is more complex to assess and more detailed than that of a one-off sport event. The analysis of a whole season encompasses many aspects, including different stadium locations and capacities, varying attendance and travel patterns, and the distribution of home and away fans within matches. The present research was also the first study of analytical nature in spectator sports as it analysed the factors that are associated with seasonal carbon footprint. Thus it goes beyond the descriptive results of previous research in this field. Moreover, this study was the first to relate the concepts of environmental consciousness and team commitment to the environmental impact of fan travel in a professional sport league.

The results of this study have practical implications for several Bundesliga stakeholders including policy makers, league officials, club authorities and fans. Understanding the carbon footprint associated with match attendance in a professional sport league is crucial for league authorities and clubs to develop evidence-based measures and promote environmentally friendly behaviour among fans. The analytical findings provide relevant knowledge about the main drivers of emissions and can guide the development of initiatives to reduce emissions. Since environmental consciousness of fans did not translate into pro-environmental behaviour, policy makers and club officials should create conditions for fans to overcome the evident environmental value-action gap. This gap can be bridged by turning high-cost situations into low-cost situations, for example by reducing the perceived costs (e.g. long waiting time, delays, overcrowding) of transport modes that cause lower emissions like trains or local public transport. Although Bundesliga clubs already established the free use of public transport to and from Bundesliga match venues in combination with a match ticket, cars are still the most frequently used mode of transport to attend matches. Thus, local authorities

need to increase the attractiveness of public transport and provide a more convenient public transport system on Bundesliga matchdays. Furthermore, Bundesliga clubs could cooperate with national railway companies to offer train ticket discounts for fans to promote long-distance rail travel. Another option is to offer team-sponsored trains or buses that encourage especially *devoted travellers* and *casual visitors* to stay away from car travel.

Since professional sport leagues can use their visibility for facilitating wider changes in society, Bundesliga clubs need to raise environmental awareness among their supporters and promote pro-environmental behaviour. This is especially relevant for committed fans who tend to be younger, less environmentally conscious and less educated, and who favour car use over public transport. Public environmental awareness campaigns educating the fan base about the carbon footprint associated with match attendance might be one way to encourage more environmentally sustainable practices (Casper et al., 2014; Cooper, 2020). Such campaigns would be an opportunity for collaboration between science and practice (McCullough, 2018). The inherent challenge for Bundesliga clubs lies in the finding that more committed fans produce higher emissions. Hence, clubs need to navigate the trade-off between emotional attachment to the club and environmental impact of their fans. Since Bundesliga authorities do not want to keep fans away from Bundesliga matches, the only way to reduce their carbon footprint is to educate them about more environmentally sustainable ways of travelling and incentivise their usage. High commitment levels might foster the participation in educational campaigns and environmental initiatives by football clubs.

The DFL as the umbrella organisation for professional football in Germany should include environmental sustainability criteria in addition to numerous economic and talent-related requirements into its licensing procedure. This would force Bundesliga clubs to implement pro-environmental strategies. Due to a general societal change towards more environmentally sustainable habits, sport organisations nowadays are expected to be active in terms of pro-environmental strategies (Casper et al., 2014). The findings of this study should also encourage football fans to reconsider their travel behaviour and use more environmentally friendly transport modes.

This study has some limitations that can guide future research. The estimation of carbon footprint was limited to the national emission factor table including only direct emissions that are based on the average degree of capacity utilisation for each mode of transport. Thus, indirect and life cycle emissions could not be included. Carpooling could not be considered either, which can, however, be a common way of travelling for fans attending football matches together. Moreover, the assessment of travel emissions assumed that fans used the same route and the same modes of transport on their way to and from the Bundesliga matches, which might not have always been the case. However, a more detailed survey capturing additional travel information might have led to a higher drop-out rate. Although the carbon footprint of fans was estimated as detailed as possible, the results may suffer from recall biases since respondents might have had difficulties to remember the exact modes of transport for every single match they attended. Future research could address these limitations and assess the carbon footprint in spectator sports by using tracking technology recording travel distances more accurately. Another limitation is that this study focused on the organisational boundary of Bundesliga matches, neglecting emissions caused by other leisure activities

that fans might choose instead of attending Bundesliga matches. Understanding the emissions caused by other leisure activities would allow comparisons with Bundesliga-related emissions and provide a more holistic picture of the emissions resulting from leisure activities.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix

Table A1. Psychological commitment to team scale (n = 539).

No.	Item (1 = *strongly disagree*; 7 = *strongly agree*)	Mean	SD	Cronbach's α
1*	I might rethink my allegiance to my favourite team if this team consistently performs poorly	6.68	0.84	0.805
2	I would watch a game featuring my favourite Bundesliga team regardless of which team they are playing	6.21	1.42	
3*	I would rethink my allegiance to my favourite team if management traded away its best players	6.61	0.98	
4	Being a fan of my favourite Bundesliga team is important to me	6.04	1.30	
5	Nothing could change my allegiance to my favourite Bundesliga team	5.72	1.59	
6	I am committed to my favourite Bundesliga team	6.32	1.05	
7	It would not affect my loyalty to my favourite Bundesliga team if management hired a head coach that I disliked very much	6.19	1.31	
8*	I could easily be persuaded to change my favourite Bundesliga team	6.83	0.83	
9	I have been a fan of my favourite team since I began watching professional football	5.75	2.05	
10	I could never switch my loyalty from my favourite Bundesliga team even if my close friends were fans of another team	6.53	1.28	
11	It would be unlikely for me to change my allegiance from my current favourite Bundesliga team to another	6.70	1.06	
12	It would be difficult to change my beliefs about my favourite Bundesliga team	6.22	1.14	
13	You can tell a lot about a person by their willingness to stick with a team that is not performing well	6.09	1.29	
14*	My commitment to my favourite Bundesliga team would decrease if they were performing poorly and there appeared little chance their performance would change	6.24	1.43	
	Fan_commitment (index)	6.30	0.69	

Note: *reverse-coded.

Table A2. Environmental consciousness scale (n = 539).

No.	Item (1 = *strongly disagree*; 5 = *strongly agree*)	Mean	SD	Cronbach's α
1	I am afraid when I think about environmental conditions for future generations	3.95	1.12	.888
2	If we continue our current style of living, we are approaching an environmental catastrophe	4.00	1.12	
3	Watching TV or reading in the newspaper about environmental problems, I am often embarrassed and angry	3.24	1.18	
4	The great majority of German people do not act in an environmentally responsible way	3.89	0.93	
5	There are limits of economic growth which the industrialised world has already reached or will reach soon	3.19	1.21	
6*	In my opinion environmental problems are greatly exaggerated by proponents of the environmental movement	3.67	1.20	
7	It is still true that politicians do much too little to protect the environment	4.14	1.06	

(*Continued*)

Table A2. Continued.

No.	Item (1 = *strongly disagree*; 5 = *strongly agree*)	Mean	SD	Cronbach's *α*
8	To protect the environment, we all should be willing to reduce our current standard of living	3.58	1.19	
9	Environmental protection measures should be carried out, even if this reduces the number of jobs in the economy	3.39	1.21	
	Env_conscious (index)	3.67	0.83	

Note: *reverse-coded.

Table A3. Emission factors and average degree of capacity utilisation by mode of transport for the reference year 2017 (Federal Environmental Office, 2018b).

Mode of transport	Emission factor (in g CO_2-e) per passenger-km	Average degree of capacity utilisation
Car	139	1.5 persons per car
Coach	32	60%
Long-distance train	36	56%
Local train	60	27%
Public bus	75	21%
Subway, tram	64	19%
Plane	201	82%